

Smart Warehouse System

Course: Database Systems | Platform: MS SQL Server

Presented by: Team Members

Introduction to SWMS

The Smart Warehouse Management System (SWMS) is an enterprise logistics tool designed to:

-  Manage products effectively
-  Track inventory precisely
-  Process orders autonomously
-  Optimize warehouse operations
-  Improve database performance

 Modern automated warehouse interior

Problem & Solution

Traditional Problems

- Slow inventory tracking
- Error-prone manual operations
- Poor order management
- Data inconsistency
- Weak security protocols
- Unoptimized, slow database queries

The Solution

SWMS fully automates critical warehouse operations using the advanced capabilities of Microsoft SQL Server.

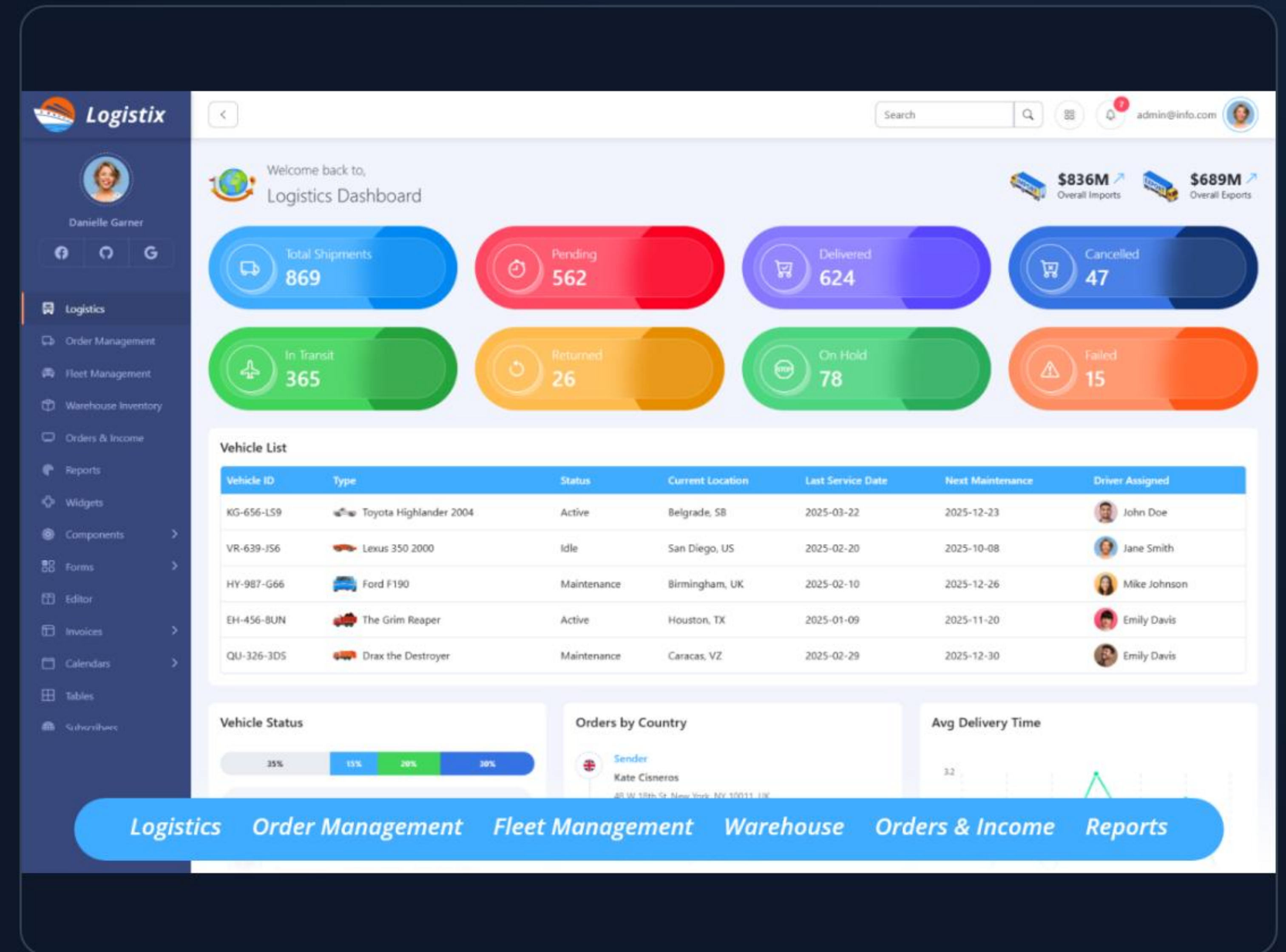
By shifting business logic directly to the database layer, it guarantees data integrity, accelerates processing, and enforces strict, role-based security.

System Departments

-  **Inbound Department:** Manages supplier relationships and registers incoming shipments.
-  **Inventory Department:** Handles internal storage locations and capacity management.
-  **Outbound Department:** Responsible for order picking, packing, and dispatch.
-  **Network Department:** Optimizes shipment batches and delivery routing.
-  **Administration:** Oversees user access, system roles, and audit monitoring.

Key Functionalities

- → Supplier Management
- → Product Catalog Management
- → Real-time Inventory Tracking
- → Automated Order Processing
- → Delivery Route Optimization
- → Immutable Audit Logging
- → Role-based Security & Permissions



Database Design

Core Entities

- **Users & Roles:** Base access control mapping.
- **Suppliers:** Source of inbound logistics.
- **Products:** Global item catalog.
- **ProductUnits:** Individual serialized item tracking within the warehouse grid.

Transactional Records

- **Orders & OrderItems:** Customer purchases and picking requirements.
- **InventoryMovements:** Historical ledger of all physical goods transfers.
- **AuditLogs:** System-wide security tracking for all DML operations.

Entity Relationship

Database Structure

The database utilizes a highly normalized schema to map the physical Warehouse Structure perfectly into the digital system.

It relies heavily on **One-to-Many Relationships** to link core entities securely.

Strongly enforced **Primary Keys** and **Foreign Keys** guarantee total referential integrity across all departments.

SQL Implementation

Technologies Used: **Microsoft SQL Server & SSMS**

-  **Tables & Constraints:** Prevent invalid data entry at the root.
-  **Relationships:** Enforced constraints linking tables logically.
-  **Views:** Abstracted data layers for secure reporting.
-  **Stored Procedures:** Encapsulated logic for faster transactions.
-  **Triggers:** Automated validations and silent auditing.

Stored Procedures

Examples

- `sp_AddSupplier`
- `sp_AddProduct`
- `sp_CreateOrder`
- `sp_AssignLocation`

Key Benefits

By moving operational logic directly into stored procedures, we achieve significantly faster execution times by reducing network traffic.

This approach provides reusable business logic and acts as a security layer against SQL injection attacks.

SQL Functions

Examples

- `fn_GetAvailableQuantity`
- `fn_GetFullLocation`
- `fn_GetCustomerOrdersCount`

Key Benefits

Functions encapsulate complex mathematical and string calculations, making them easily reusable across multiple queries.

This results in much cleaner SQL queries, accelerating development and simplifying long-term database maintenance.

Database Triggers

Implemented Triggers

- **Audit Trigger:** Records every DML change.
- **Login Trigger:** Monitors server access.
- **Delete Prevent Trigger:** Blocks DB dropping.
- **Capacity Trigger:** Validates bin limits.

Primary Purpose

Triggers provide powerful background automation, enforcing complex constraints invisibly.

They act as the ultimate fail-safe for system security and maintain a strict, automated auditing trail for compliance.

Database Views

Examples

- `vw_CurrentInventory`
- `vw_PendingOrders`
- `vw_TopProducts`

Key Benefits

- **Simplified Reporting:** Masks complex multi-table JOINS from reporting software.
- **Better Security:** Restricts user access to specific rows and columns.
- **Faster Access:** Pre-compiled query structures.

Security & Access

Role-Based Controls

-  Admin
-  InventoryManager
-  Receiver
-  Picker
-  NetworkManager

Security is reinforced through strict login auditing and heavily restricted permissions based entirely on user roles.

 Cybersecurity digital padlock secure access tech

System Auditing

The Audit Logs immutably record:

 **Username:** Identifying the user involved.

 **Operation:** Noting the action taken.

 **Table Name / Object Type:** What was affected.

 **Operation Date:** Precise timestamp tracking.

 **Login Tracking:** Tracks all database login events to catch unauthorized access attempts.

Big Data: 1M+ Records



Simulated Records

Large-Scale Dataset

To ensure enterprise readiness, the database was populated with a massive volume of data:

- 1,000,000+ Orders
- 1,000,000+ OrderItems
- 100,000 Customers

Purpose: Rigorous performance tuning and scalability testing.

Index Optimization

Before Index

Execution Plan: Table Scan

When searching through millions of records without an index, SQL Server is forced to scan every single row sequentially.

This results in slow query execution and heavy load.

After Index

Execution Plan: Index Seek

By implementing targeted Indexes, the database uses a B-Tree structure to jump directly to the required data pointers.

This delivers lightning-fast performance.

Performance Results

Query Operation	Before Index	After Index	Status
Orders by CustomerID	8,500 ms	120 ms	Highly Optimized
Search by SKU	4,000 ms	50 ms	Highly Optimized

System Reports

-  **Inventory Report:** Aggregated stock levels per item.
-  **Top Customers:** Identified by total order volume.
-  **Top Products:** Best-selling items based on frequency.
-  **Pending Orders:** Orders queued for picking.
-  **Employee Performance:** Metrics per assigned picker.

Backup & Restore

“ BACKUP DATABASE SmartWarehouseDB;

RESTORE DATABASE SmartWarehouseDB; ”

— Standard Disaster Recovery Protocol

System Advantages



Enterprise Design

Built with a highly scalable database architecture capable of handling millions of records smoothly.



Secure Architecture

Enforces role-based access control and comprehensive audit tracking for transparency.



High Performance

Heavily optimized queries and precise indexing strategies drastically reduce execution times.

Project Conclusion

SWMS successfully demonstrates:

-  Advanced SQL Server concepts.
-  Enterprise database architecture.
-  Deep performance optimization.
-  Rigorous security implementations.
-  Real-world warehouse simulation.

 Business professionals modern tech success handshake

Thank You

Questions? 

Image Sources

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